

CSMFAB Materials Study — Part V

Critical Minerals, Rare Earth Elements, and Strategic Supply Chain Analysis

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: Critical mineral supply chains for Indium, Yttrium, Cobalt, Niobium, Tungsten, Samarium, Titanium, Lithium, and Manganese across the CSMFAB portfolio. Emphasis on: supply risk, domestic alternatives, recycled sourcing, and geopolitical resilience.

Overview: Why Critical Minerals Matter for CSMFAB

Of the 30 distinct material classes in the CSMFAB portfolio, **9 involve critical or strategic minerals** either directly (as elemental ingredient) or as irreplaceable precursor. The US Geological Survey defines “critical minerals” as those essential to economic/national security with supply chains vulnerable to disruption. In 2026, China controls **dominant share of processing** for 7 of the 9 CSMFAB critical minerals — this is the portfolio’s primary geopolitical risk.

CSMFAB Critical Minerals Dependency Map:

Mineral	Products Affected	China Processing Share (2026)	Supply Risk
Indium (In)	YInMn Blue (35 products)	58% of global supply	CRITICAL
Yttrium (Y)	YSZ, YInMn Blue, Si3N4	85% of global supply	CRITICAL
Cobalt (Co)	CoAl2O4, LiFePO4 (minor), SmCo	73% processing (from DRC ore)	HIGH
Niobium (Nb)	KNbO3-BaTiO3 piezoelectric	Brazil 92% of mining	MEDIUM
Tungsten (W)	WO3 electrochromic	82% of global mining	HIGH
Samarium (Sm)	SmCo magnets	90% processing	HIGH
Titanium (Ti)	MXene, PEEK, TiO2	49% sponge production	MEDIUM
Lithium (Li)	LiFePO4 batteries	65% refining	MEDIUM
Manganese (Mn)	YInMn, CoAl2O4	36%	LOW

Critical Mineral 1 — Indium (In)

Atomic number: 49 | **Annual world production:** ~700 metric tons (2026) | **CSMFAB use:** YInMn Blue pigment

Supply Chain Deep Dive

How Indium occurs in nature: Indium is a trace element in sphalerite (zinc sulfide) ores — typically 1-200 ppm by weight. It is **never mined as a primary product** — always as a by-product of zinc smelting. Global zinc production: ~13 million tons/year □ ~700 tons In recovered as by-product.

Major production steps: 1. Zinc ore mined (galena, sphalerite): China, Peru, Australia, USA (Doe Run, Missouri) 2. Ore smelted to crude zinc □ indium concentrated in flue dusts (0.5-5% In) 3. Hydrometallurgical recovery: leach flue dust □ SX with D2EHPA or ionic liquid □ plating 4. Refined indium: 99.99% purity

Primary global suppliers: - Freeport-McMoRan (USA — from zinc operations in Arizona): ~15 t/yr - Doe Run (Missouri, USA): ~8 t/yr — US domestic - Korea Zinc (South Korea): ~180 t/yr — largest single producer - Umicore (Belgium): ~60 t/yr (recycling focus) - Various Chinese smelters: ~400 t/yr combined

Price history and volatility:

2001: \$97/kg
2006: \$975/kg (LCD boom)
2010: \$600/kg
2016: \$167/kg (oversupply)
2022: \$240/kg
2024: \$320/kg
2026: \$290-380/kg (China export control uncertainty)

Domestic US Indium Recovery Strategy

Critical action: CSM should position as an indium recycler, not just consumer.

US-domestic indium recovery sources: 1. Zinc smelter by-product streams (Missouri, Tennessee): negotiate off-stream agreement 2. Liquid crystal display (LCD) recycling: ITOs contain ~0.5 kg In/kg coating 3. CSM Phoenix Protocol In recovery from retired Aegis products

Indium from LCD glass recycling: - US LCDs discarded per year: ~350 million units - In content per TV: 0.5–2g - Recovery potential: 50–100 tons/yr (currently mostly landfilled) - CSM opportunity: partner with electronics recycler □ capture In stream

Estimated in-recovery cost: \$18–\$35/kg (vs. \$290–\$380 virgin mine price) **This makes YInMn Blue recycling the most financially compelling Phoenix Protocol component.**

Indium Risk Mitigation Strategies (ranked by feasibility)

1. **Indium-sparing YInMn formula** — reduce In content 40-60% via Ga/Fe/Al substitution (Priority R&D: ~18-month development program; cost ~\$280,000)
2. **Phoenix Protocol ionic liquid recovery** — recover 99.4% from each retired panel (Immediate action; requires ionic liquid solvent procurement)
3. **Domestic co-product agreement** with Doe Run (Missouri) or Freeport zinc operations (Negotiate 3-year supply contract; volume commitment for preferred pricing)
4. **CoAl₂O₄ substitution** for applications where Indium-free alternative acceptable (70-80% NIR performance vs. 85-92% for YInMn — trade-off decision per product)
5. **Strategic stockpile** — purchase 12-24 months' supply at favorable price periods (Working capital: \$1.2M buys ~4 tons; covers portfolio production for 2+ years)

Critical Mineral 2 — Yttrium (Y)

Atomic number: 39 | **Class:** Rare Earth Element (light REE group, but often processed with HREE) | **CSMFAB use:** 3Y-TZP zirconia, YInMn Blue, Si₃N₄ sintering aid

Supply Chain

Occurrence: Yttrium is found in bastnäsite and monazite REE ores (alongside La, Ce, Pr, Nd). Also in xenotime (yttrium-dominant REE phosphate ore) — more unusual.

Global production: - China: ~85% of REE mining and essentially all REE separation - Mountain Pass (California, MP Materials): REO concentrate exported to China for separation - Lynas Rare Earths (Australia/Malaysia): produces separated RE oxides without China - Energy Fuels (Utah): building US heavy REE separation capability (2024–2026)

Yttrium oxide (Y₂O₃) pricing: - 2020: \$4.50/kg - 2022: \$12/kg (REE price spike) - 2024: \$35–\$50/kg (heavy REE tightening) - 2026: \$38–\$55/kg

US Domestic Yttrium Sources

Bear Lodge (Wyoming): Nearest-term large US heavy REE deposit with significant Y content. Rare Element Resources — awaiting final permitting.

Pea Ridge (Missouri): Iron ore deposit with recoverable HREE including Y — advanced drilling.

Round Top (Texas): Texas Mineral Resources — large peralkaline deposit; 2% Y₂O₃ in deposit. Mining schedule: 2027-2028 production begins (announced).

Energy Fuels (White Mesa Mill, Utah): Processing monazite sand □ separating Y₂O₃. Current production: ~5 tons Y/yr; target 50 tons/yr by 2027.

Yttrium recovery from Phoenix Protocol: From retired YSZ ceramic parts: HCl dissolution □ selective extraction with Cyanex 272 (bis(2,4,4-trimethylpentyl)phosphinic acid) □ 98% Y₂O₃ recovery. In-recovery cost: \$22–\$32/kg vs. \$38–\$55/kg virgin.

Critical Mineral 3 — Cobalt (Co)

Atomic number: 27 | **Annual world production:** ~200,000 mt | **CSMFAB use:** CoAl₂O₄ pigment, SmCo magnets, LiFePO₄ (minor)

Supply Chain (Dominated by DRC)

Democratic Republic of Congo (DRC) situation: - 69% of world cobalt mining (Katanga region) - Artisanal small-scale mining: 30% of DRC output, significant humanitarian concerns - Major mine operators: Glencore (Swiss), CMOC (Chinese), Ivanhoe Mines (Canadian)

Price history: - 2018: \$95,000/mt (electric vehicle boom panic) - 2020: \$38,000/mt - 2022: \$82,000/mt - 2024: \$28,000/mt (oversupply from DRC expansion) - 2026: \$32,000–\$45,000/mt (normalized range)

CSMFAB Cobalt Demand Analysis

CoAl₂O₄ cobalt content: ~14.5 wt% Co per kg CoAl₂O₄ Annual portfolio demand (at scale): ~220 kg Co/yr □ ~1.5 metric ton CoAl₂O₄/yr Annual Co cost: 220 kg × \$35/kg = **\$7,700/yr** — manageable

Key insight: CoAl₂O₄ uses very little cobalt relative to battery industry (LCo cathodes). CSMFAB is NOT at risk from cobalt supply despite DRC concerns.

US Domestic Cobalt Sources

- **Idaho Cobalt Belt:** Jervois Global (Australia/USA) — Idaho Cobalt Operations Opened 2023; first US primary cobalt mine in 60 years. 1,500 t/yr capacity.
- **Midwest cobalt belt (Missouri):** Cobalt in copper-cobalt sulfide deposits (Viburnum Trend)
- **Alaska:** Several Co-Ni deposits under exploration

LiFePO₄ batteries (CSMFAB product 050, 019): Use Fe and P, not Co cathode □ **cobalt-free battery chemistry** already chosen. Smart decision.

Critical Mineral 4 — Niobium (Nb)

Atomic number: 41 | **Annual world production:** ~75,000 mt (Nb₂O₅) | **CSMFAB use:** KNbO₃-BaTiO₃ piezoelectric ceramics

Supply Chain (Near-Monopoly: Brazil)

Brazil dominates niobium production at 92%: - CBMM (Companhia Brasileira de Metalurgia e Mineração): 75% of world Nb supply from single mine (Araxá) - A single Brazilian company controls the majority of world supply of a critical material

Price (2026): Nb₂O₅: \$32–\$55/kg | Ferroniobium (steel industry): \$36–\$42/kg Nb basis

US domestic niobium: - Elk Creek (Nebraska): NioCorp Developments — NbC/Nb₂O₅ production target 2027 - Granite-hosted pegmatite deposits in several states — under-explored

Risk level for CSMFAB: MEDIUM — Brazil is a stable ally; no current supply disruption. Single-source concentration is the primary risk. Mitigation: 12-month strategic inventory.

In-house Nb recovery: Not practical at current portfolio scale.

Critical Mineral 5 — Tungsten (W)

Atomic number: 74 | **Density:** 19.3 g/cm³ (densest practical metal) | **CSMFAB use:** WO₃ electrochromic, WC spheres for LRAM

Supply Chain (China #1 at 82%)

Global production distribution: - China: 72,000 mt/yr (82% of world) - Vietnam: 4,500 mt/yr - Russia: 2,000 mt/yr - Portugal: 1,800 mt/yr (Panasqueira mine — Europe's significant W producer) - USA: currently minimal (tungsten mining ceased; net importer)

Price volatility: - APT (Ammonium Paratungstate) industrial standard - 2022-2026 range: \$280-\$375/metric ton unit (MTU) - 2026: \$340/MTU = \$34/kg W

US domestic recovery: - US was #1 W producer pre-1980s; mineral rights exist in Nevada, Idaho, Colorado - Almonty Industries (Canada) — operating Sangdong mine (South Korea); North American assets - Proposed Nevada mines: under EIS review 2026

CSMFAB W demand: Small — primarily WO₃ for window film and WC spheres for acoustic dampers. Total annual demand (at scale): ~180 kg W equivalent. **Strategic stock of 500 kg APT (\$170 cost) covers 2+ years. Recommended.**

Critical Mineral 6 — Samarium (Sm) — SmCo Magnets

CSMFAB use: SmCo permanent magnets in motor electromagnets (product 029, 035 control coils) **NOT LiFePO₄ — SmCo is used only in small control magnets, not primary propulsion**

Properties of SmCo Magnets

- BH_{max}: 20–32 MGOe (excellent performance)
- Max operating temperature: 350°C (far exceeds NdFeB's 150°C)
- Price resistance: does not degrade in radiation (unlike NdFeB which can be bulk-demagnetized by GIC)
- **This is why SmCo is preferred in Aegis products over NdFeB — thermal margin for fire scenarios**

Acquisition

- Electron Energy Corporation (Landisville, PA): **US domestic SmCo manufacturer** — key strategic supplier
- Arnold Magnetic Technologies (Rochester, NY): US-domestic rare earth magnet manufacturer
- TDK Magnetics (US facility): NdFeB focus but some SmCo capability
- Price: \$75–\$120/kg (Sm2Co17 grade B alloy)

Critical action: Establish exclusive supply relationship with Electron Energy Corporation before competition. They are the only significant US domestic SmCo producer — a strategic national asset.

Critical Mineral 7 — Titanium (Ti)

CSMFAB use: Primary in MXene precursor (Ti3AlC2), secondary in TiO2 components

Global Supply and Domestic Resources

Mineral sources: - Ilmenite (FeTiO3): Abundant globally — Australia, South Africa, Canada, USA (Virginia Beach heavy mineral sands) - Rutile (TiO2): Rarer, purer — Australia, Sierra Leone, Mozambique, USA (Virginia)

US domestic titanium: - **Virginia Heavy Minerals (VHM):** Major ilmenite/rutile reserve in Virginia Beach area - **The Chemours Company (Savannah, GA):** Produces TiO2 from domestic minerals - **Ti-rich basalt:** Eastern US basalt provinces contain recoverable Ti (synergy with basalt fiber program)

Kroll Process (Ti metal from TiCl4): $\text{TiO}_2 + 2\text{Cl}_2 + 2\text{C} \rightarrow \text{TiCl}_4 + 2\text{CO}$ $\text{TiCl}_4 + 2\text{Mg} \rightarrow \text{Ti} + 2\text{MgCl}_2$ Primary Kroll suppliers: TIMET (Titanium Metals Corp, Texas), Allegheny Technologies

Price 2026: TiO2 \$3–\$8/kg; Ti sponge \$18–\$28/kg; Ti powder (attrition milled) \$45–\$120/kg

Consolidated Critical Mineral Risk Register

Mineral	Risk Level	Primary Risk	Principal Mitigation
Indium	CRITICAL	China supply concentration + volatility	Phoenix Protocol recycling + In-sparing formula R&D + Doe Run agreement
Yttrium	CRITICAL	85% China processing	Energy Fuels supply agreement + Y recovery from scrap
Cobalt	HIGH	DRC instability + ethical concerns	Idaho Cobalt Operations domestic supply; LiFePO4 avoids Co in batteries
Tungsten	HIGH	82% China mining	Strategic APT stockpile; Portugal (Panasqueira) backup; WC recovery from LRAM parts
Samarium	HIGH	90% China RE processing	Electron Energy Corp (US domestic SmCo) exclusive agreement

Mineral	Risk Level	Primary Risk	Principal Mitigation
Niobium	MEDIUM	Brazil 92% concentration	12-month strategic inventory; NioCorp Elk Creek (2027)
Titanium	MEDIUM	Processing concentration	Virginia heavy mineral sands domestic; TIMET domestic supply
Lithium	MEDIUM	65% China refining	Energy Fuels (UT) + Piedmont Lithium (NC) + Round Top (TX)
Manganese	LOW	36% China	Multiple global sources; Consol Energy domestic coal Mn by-product

CSMFAB Critical Minerals Procurement Strategy

Phase 1 (Immediate, 0-6 months):

1. Sign supply agreement with Doe Run (Missouri) for co-product In stream
2. Sign supply agreement with Electron Energy Corp (PA) for SmCo magnets
3. Establish Shepherd Color Co. volume contract for YInMn Blue (lock 3-year pricing)
4. Begin Phoenix Protocol ionic liquid In/Y recovery pilot (equipment purchase \$45,000)
5. Purchase 500 kg APT tungsten strategic stockpile (\$170,000)

Phase 2 (6-18 months):

1. Fund Indium-sparing YInMn formula R&D (\$280,000 program)
2. Establish Energy Fuels (Utah) Y₂O₃ supply relationship for when their production scales
3. In-house KNbO₃-BaTiO₃ synthesis program launch (shares ceramics lab with ZrB₂ program)
4. Identify and sample US domestic heavy mineral ilmenite for possible Ti in-house sourcing

Phase 3 (18-48 months):

1. Full in-house YInMn synthesis with indium-sparing formula (after R&D success)
2. Begin in-house MXene synthesis from in-house Ti₃AlC₂ MAX Phase
3. License Cyanex 272 Y-extraction process for expanded Phoenix Protocol program
4. Evaluate Round Top (TX) peralkaline ore for portfolio supply once in production (2028)

End of CSMFAB Materials Study Part V | Critical Minerals and Rare Earth Supply Chain Carrington Storm Motors | June 2026